Guide Review 1 (7 Marks)

Research-Based Project

Expected Outcome: The student should have finalized the research problem, designed the complete methodology, and initiated implementation.

Component	Marks	Rubrics
Problem Statement & Research Gap Refinement	**1**	**0:** Problem statement is unclear or inconsistent with Phase I findings. **1:** Problem statement, objectives, and research gap are clearly refined and technically justified.
Proposed Methodology & Solution Design	**3**	**0:** No methodology or workflow prepared. **2:** An architecture/workflow is presented but lacks sufficient technical detail or justification. **3:** Complete methodology including workflow, architecture, algorithms/mathematical model, datasets, implementation plan, and evaluation strategy with proper technical justification.
Initial Implementation Progress	**2**	**0:** Implementation not initiated. **1:** Development environment prepared with initial coding or preprocessing completed. **2:** Initial implementation (30+% of completeness) demonstrates meaningful progress through code, dataset preparation, baseline implementation, or prototype modules.
Regularity & Technical Involvement	**1**	**0:** Irregular meetings or inactive participation. **1:** Regular meetings with active technical discussions and continuous progress.

Application-Based Project

Expected Outcome: The student should have finalized the application requirements, completed the system design, and initiated implementation.

Component	Marks	Rubrics
Problem Definition & User Requirement Analysis	**1**	**0:** Problem statement or user requirements are unclear or incomplete. **1:** Problem statement, functional requirements, non-functional requirements, target users, and project objectives are clearly defined and justified.
Application Architecture & Solution Design	**3**	**0:** No application architecture prepared. **2:** A workflow or architecture available but incomplete. **3:** Complete architecture including workflow, module interaction, technology stack, and deployment approach with proper justification.
Initial Implementation Progress	**2**	**0:** Development not started. **1:** Initial setup completed with one or two functional modules implemented. **2:** Initial application modules, with implementation evidence (GitHub, prototype, screenshots, etc.) amounting to 30+% of overall.
Regularity & Technical Involvement	**1**	**0:** Irregular meetings or poor participation. **1:** Regular interaction with the guide and active technical discussions throughout the review period.

Software-Based Project

Expected Outcome: The student should have finalized the software requirements, completed the software architecture, and initiated software development.

Component	Marks	Rubrics
Software Requirements Specification (SRS)	**1**	**0:** Functional and non-functional requirements are incomplete or unclear. **1:** Well-defined Software Requirements Specification (SRS) including functional requirements, non-functional requirements, use cases, constraints, and project scope.
Software Architecture & System Design	**3**	**0:** No architecture or system design prepared. **2:** Basic architecture presented but lacks clarity in module interaction. **3:** Complete software architecture, including UML diagrams, module decomposition, database schema, API specifications, UI workflow, technology stack, and deployment strategy with proper justification.
Initial Development Progress	**2**	**0:** Software development not initiated. **1:** Initial implementation of backend, frontend, or database completed. **2:** Initial software modules are functional with implementation evidence such as GitHub commits, working interfaces, APIs, database connectivity, or authentication modules.
Regularity & Technical Involvement	**1**	**0:** Irregular meetings or inactive participation. **1:** Regular meetings, active technical discussions, continuous GitHub activity, and consistent progress.

Panel Review 1 (20 Marks)

Research-Based Project

Expected Outcome: The student should have completed the methodology design and 60+% implementation, and the major modules should be functional.

Component	Marks	Rubrics
Methodology & Solution Design	**5**	**0:** Methodology incomplete or technically weak. **3:** Reasonable methodology but lacks sufficient justification or workflow clarity. **5:** Complete methodology including workflow, architecture, algorithms/mathematical model, datasets, evaluation metrics, and implementation strategy with proper technical justification.
Implementation Progress (≈60%)	**8**	**0:** Less than 30% implementation completed. **4:** Approximately 30–40% of the proposed implementation completed with partial implementation of the proposed methodology and limited module integration. **6:** Approximately 40–50% of the proposed implementation completed for major modules and individually functional; however, integration between modules is incomplete and the overall solution is only partially operational **8:** Approximately 60% of the proposed implementation completed for major modules, integrated, and the core functionality of the proposed solution is successfully demonstrated.
Technical Knowledge	**5**	**0:** Unable to explain methodology or implementation. **3:** Satisfactory understanding with minor technical gaps. **5:** Excellent understanding of the proposed methodology, implementation, and design choices.
Presentation & Communication Skills	**2**	**0**: Presentation is poorly organized; unable to communicate the work effectively. **1**: Presentation is satisfactory but lacks clarity, confidence, or proper organization. **2**: Professional, well-organized presentation with clear communication, effective use of presentation materials, proper time management, and confident responses to questions.

Application-Based Project

Expected Outcome: The student should complete the application architecture, implement 60%+ of the application, and demonstrate the major modules.

Component	Marks	Rubrics
Application Architecture & Workflow Design	5	**0:** Architecture incomplete or poorly designed. **3:** Basic architecture available but module interaction requires improvement. **5:** Complete architecture including workflow, module interaction, UI flow, database schema, APIs/services, technology stack, and deployment strategy.
Implementation Progress (≈60%)	**8**	**0:** Less than 30% implementation completed. **4:** Approximately 30–40% of the proposed implementation completed; core modules have been initiated, but functionality and integration are limited. **6:** Approximately 40–50% of the proposed implementation completed; most major modules are implemented with partial integration and demonstration of core functionalities. **8:** Approximately 60% of the proposed implementation completed; major modules are functional, partially integrated, and successfully demonstrated.
Technical Knowledge	**5**	**0:** Poor understanding of application architecture and technologies. **3:** Satisfactory understanding of implementation and workflow. **5:** Strong technical knowledge with confident demonstration of application architecture, implementation, and design decisions.
Presentation & Communication Skills	**2**	**0**: Presentation is poorly organized; unable to communicate the work effectively. **1:** Presentation is satisfactory but lacks clarity, confidence, or proper organization. **2**: Professional, well-organized presentation with clear communication, effective use of presentation materials, proper time management, and confident responses to questions.

Software-Based Project

Expected Outcome: The student should complete software architecture, implement approximately 60% of the software, and integrate and demonstrate the core modules.

Component	Marks	Rubrics
Software Architecture & Module Design	**5**	**0:** Architecture incomplete or lacks module definition. **3:** Basic architecture available but APIs/database/module interaction require improvement. **5:** Complete architecture including UML diagrams, database schema, APIs, module interaction, UI workflow, deployment architecture, and technology stack.
Implementation Progress (≈60%)	**8**	**0:** Less than 30% implementation completed. **4:** Approximately 30–40% of the proposed implementation completed. Core software modules (backend, frontend, or database) have been initiated, but functionality and integration are limited. **6:** Approximately 40–50% of the proposed implementation completed. Most major software modules are functional; however, integration between the backend, frontend, database, and APIs is only partial, and the overall system is not fully operational. **8:** Approximately 60% implementation completed with backend, frontend, database, and core modules functioning and demonstrated successfully.
Technical Knowledge	**5**	**0:** Poor understanding of software architecture and implementation. **3:** Satisfactory technical understanding with minor gaps. **5:** Excellent understanding of software architecture, implementation, integration, and technical decisions with confident presentation.
Presentation & Communication Skills	**2**	**0**: Presentation is poorly organized; unable to communicate the work effectively. **1**: Presentation is satisfactory but lacks clarity, confidence, or proper organization. **2**: Professional, well-organized presentation with clear communication, effective use of presentation materials, proper time management, and confident responses to questions.

Guide Review 2 (8 Marks)

Research-Based Project

Expected Outcome: Students should have completed 85–90% of the proposed implementation, integrated the major modules, and obtained preliminary results.

Component	Marks	Rubrics
Implementation Progress	**4**	**0:** Minimal or no progress since Panel Review 1. **2:** Around 50–70% implementation completed; some major modules still incomplete. **4:** Approximately 85–90% of the proposed methodology implemented with major modules integrated and functioning.
Preliminary Verification & Results	**2**	**0:** No testing or verification performed. **1:** Initial experiments/testing performed with limited observations. **2:** Preliminary evaluation demonstrates that the proposed solution functions correctly and produces meaningful initial results.
Regularity & Technical Involvement	**2**	**0:** Irregular meetings or inactive participation. **1:** Regular meetings with moderate participation. **2:** Consistent interaction with the guide and active technical discussions throughout the semester.

Application-Based Project

Expected Outcome: Students should have completed 85–90% of the application, integrated the major modules, and demonstrated a stable prototype.

Component	Marks	Rubrics
Implementation Progress	**4**	**0:** Minimal progress since Panel Review 1. **2:** Around 50–70% of the application completed with partial module integration. **4:** Approximately 85–90% of the application implemented with major functional modules integrated successfully.
Module Integration & Functional Testing	**2**	**0:** Modules not integrated or tested. **1:** Partial integration with basic functional testing. **2:** Major modules integrated successfully with functional testing demonstrating expected application behaviour.
Regularity & Technical Involvement	**2**	**0:** Irregular meetings or inactive participation. 1: Regular meetings with moderate technical discussions. **2:** Consistent interaction with the guide and active contribution throughout the semester.

Software-Based Project

Expected Outcome: Students should have completed 85–90% of the software system, integrated the backend, frontend, and database, and performed initial testing.

Component	Marks	Rubrics
Implementation Progress	**4**	**0:** Minimal progress since Panel Review 1. **2:** Around 50–70% implementation completed with partial module integration. **4:** Approximately 85–90% of the software system implemented with major modules integrated successfully.
System Integration & Initial Testing	**2**	**0:** Backend, frontend, and database not integrated. **1:** Partial integration with basic testing completed. **2:** Backend, frontend, database, and APIs integrated successfully with initial functional testing demonstrating expected system behaviour.
Regularity & Technical Involvement	**2**	**0:** Irregular meetings or inactive participation. **1:** Regular meetings with moderate participation. **2:** Consistent interaction with the guide and active technical discussions throughout the semester.